# Introduction to GAMs with mgcv

GAM.jl Contributors

## Table of contents



### Overview

A **Generalized Additive Model (GAM)** extends the linear model by replacing linear predictor terms with smooth functions of covariates. The model takes the form:

$$g(\mu_i) = \beta_0 + f_1(x_{1i}) + f_2(x_{2i}) + \cdots + f_p(x_{pi})$$

where $g$ is a link function, $\mu_i = \mathbb{E}(y_i|\mathbf{x}_i)$, and each $f_j$ is a smooth function estimated from the data using penalized regression splines.

The smooth functions are represented as linear combinations of basis functions:

$$f_j(x) = \sum_{k=1}^{K_j} \beta_{jk}\, b_{jk}(x)$$

Smoothness is controlled by a wiggliness penalty, and the smoothing parameter $\lambda$ is estimated automatically (e.g., by REML).

This vignette demonstrates fitting a simple GAM to simulated data using **mgcv** in R.

## Setup

```
library(mgcv)
```

```
Loading required package: nlme
```

```
This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.
```

```
library(gratia)
library(ggplot2)
```

## Simulating data

We simulate $n = 200$ observations from a sine curve with Gaussian noise:

$$y_i = \sin(2\pi x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 0.3^2)$$

```
set.seed(42)
n <- 200
x <- sort(runif(n))
y <- sin(2 * pi * x) + 0.3 * rnorm(n)
df <- data.frame(x = x, y = y)
head(df)
```

```
             x           y
1 0.0002388966  0.36179064
2 0.0013808436  0.32210131
3 0.0015705542 -0.29109467
4 0.0022729661  0.56882555
5 0.0039483388 -0.17522642
6 0.0073341469  0.07771964
```

## Fitting a GAM

We fit a GAM with a cubic regression spline (`bs = "cr"`) smooth of `x` using 15 basis functions:

```r
m <- gam(y ~ s(x, k = 15, bs = "cr"), data = df, method = "REML")
summary(m)
```

```
Family: gaussian
Link function: identity

Formula:
y ~ s(x, k = 15, bs = "cr")

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.1002     0.0205  -4.888 2.15e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
       edf Ref.df     F p-value
s(x) 7.722  9.369 122.5  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.852   Deviance explained = 85.8%
-REML = 49.744  Scale est. = 0.08403   n = 200
```

The model summary shows parametric coefficients, smooth term EDF, deviance explained, and scale estimate.

### Coefficients and deviance explained

```r
cat("Number of observations:", nobs(m), "\n")
```

```
Number of observations: 200
```

```r
cat("EDF per smooth:        ", round(pen.edf(m), 2), "\n")
```

```
EDF per smooth:         7.72
```

```r
cat("Deviance explained:    ", round(summary(m)$dev.expl * 100, 1), "%\n")
```

```
Deviance explained:     85.8 %
```

```r
cat("Scale estimate:        ", round(m$scale, 4), "\n")
```

```
Scale estimate:         0.084
```

### Smooth estimates

The `gratia::smooth_estimates()` function evaluates the estimated smooth on a regular grid and returns pointwise standard errors:

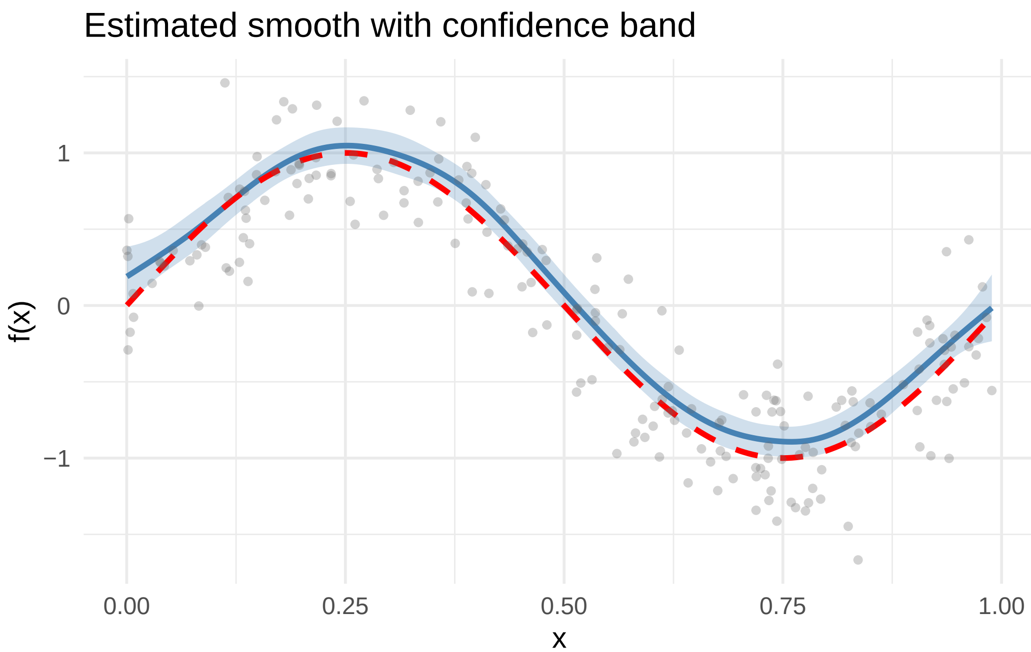
```r
se <- smooth_estimates(m, n = 200)
head(se)
```

```
# A tibble: 6 x 6
  .smooth .type .by   .estimate    .se        x
  <chr>   <chr> <chr>     <dbl>  <dbl>    <dbl>
1 s(x)    CRS   <NA>      0.190 0.0976 0.000239
2 s(x)    CRS   <NA>      0.208 0.0915 0.00521
3 s(x)    CRS   <NA>      0.226 0.0859 0.0102
4 s(x)    CRS   <NA>      0.244 0.0809 0.0151
5 s(x)    CRS   <NA>      0.263 0.0765 0.0201
6 s(x)    CRS   <NA>      0.281 0.0729 0.0251
```

We plot the smooth estimate with ±2 SE confidence bands, overlaying the true function:

```r
ggplot(se, aes(x = x, y = .estimate)) +
  geom_ribbon(aes(ymin = .estimate - 2 * .se, ymax = .estimate + 2 * .se),
              alpha = 0.25, fill = "steelblue") +
  geom_line(linewidth = 1, colour = "steelblue") +
  geom_line(aes(y = sin(2 * pi * x)), linetype = "dashed",
            linewidth = 1, colour = "red") +
```

```
  geom_point(data = df, aes(x = x, y = y), alpha = 0.3, size = 1, colour = "grey40") +
  labs(x = "x", y = "f(x)", title = "Estimated smooth with confidence band") +
  theme_minimal()
```



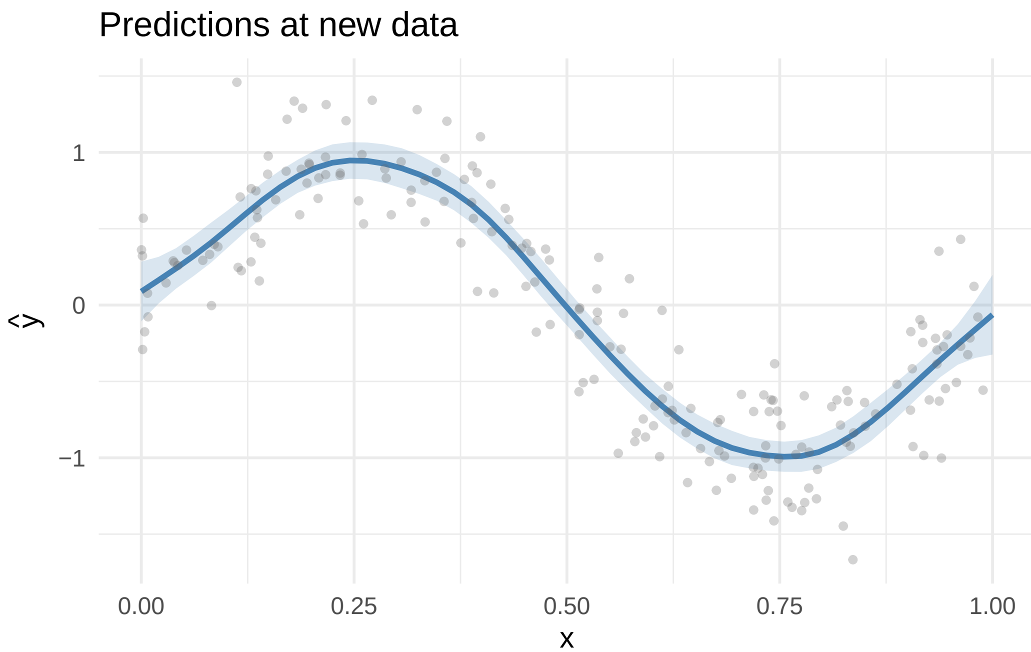

#### Prediction at new points

We can predict at new covariate values with standard errors:

```
x_new <- data.frame(x = seq(0, 1, length.out = 50))
pred <- predict(m, newdata = x_new, type = "response", se.fit = TRUE)

x_new$fit <- pred$fit
x_new$se <- pred$se.fit

ggplot(x_new, aes(x = x, y = fit)) +
  geom_ribbon(aes(ymin = fit - 2 * se, ymax = fit + 2 * se),
              alpha = 0.2, fill = "steelblue") +
  geom_line(linewidth = 1, colour = "steelblue") +
  geom_point(data = df, aes(x = x, y = y), alpha = 0.3, size = 1, colour = "grey40") +
  labs(x = "x", y = expression(hat(y)), title = "Predictions at new data") +
  theme_minimal()
```




### Smoothing parameters and REML

mgcv estimates the smoothing parameter $\lambda$ by minimizing the Restricted Maximum Likelihood (REML) criterion by default. REML is generally preferred over GCV because it:

- Avoids the tendency to undersmooth
- Provides more stable estimates
- Has a natural Bayesian interpretation

```r
cat("Estimation method:", m$method, "\n")
```

```
Estimation method: REML
```

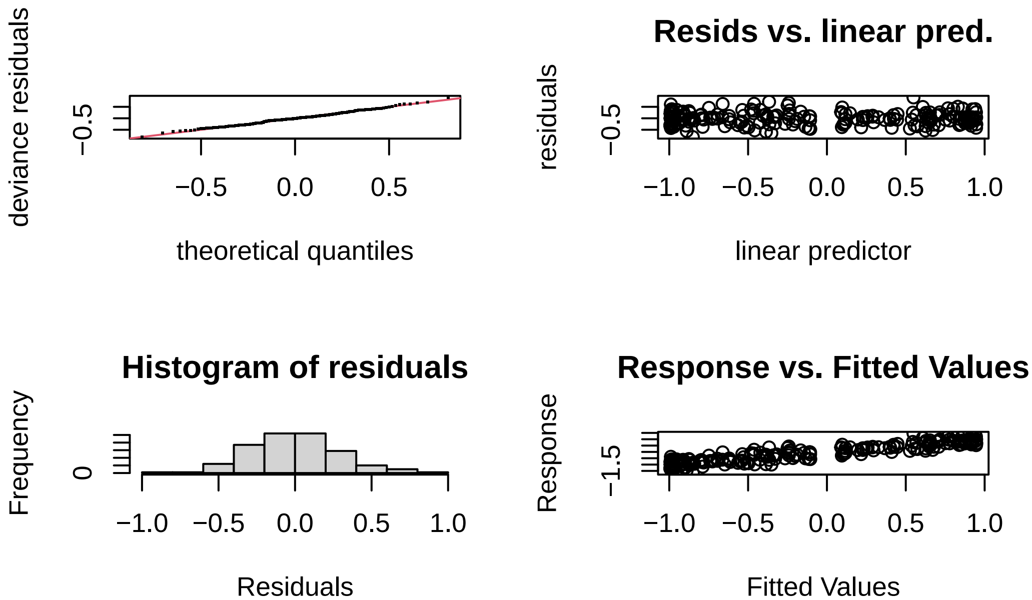
```r
cat("Smoothing parameter (log):", log(m$sp), "\n")
```

```
Smoothing parameter (log): 4.278883
```

### Model diagnostics

The `gam.check()` function provides diagnostic plots and basis dimension checks:

```r
par(mfrow = c(2, 2))
gam.check(m)
```




```
Method: REML   Optimizer: outer newton
full convergence after 4 iterations.
Gradient range [-6.776304e-10,3.284484e-10]
(score 49.74438 & scale 0.08402973).
Hessian positive definite, eigenvalue range [3.328508,99.11809].
Model rank =  15 / 15

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

        k'   edf k-index p-value
s(x) 14.00  7.72    1.09    0.88
```

```r
par(mfrow = c(1, 1))
```

## Summary

In this vignette we:

1. Simulated data from a sine curve with Gaussian noise
2. Fitted a GAM using a cubic regression spline with `k = 15` basis functions
3. Examined the model summary, EDF, and deviance explained
4. Visualized the smooth estimate with confidence bands using gratia
5. Predicted at new data points with standard errors
6. Discussed REML-based smoothing parameter estimation

The next vignette compares different smooth basis types.